Electronic supplementary material for

"Antisocial but inconsequential: large language model agents punish cooperators yet neither sustain cooperation nor reproduce its cross-cultural geography"

This supplement reports (i) the computational and reproducibility details, (ii) the full statistical tables underlying the main text, including effect sizes, confidence intervals, the complete deviation-binned punishment, the per-society breakdown and the dispositional analysis, and (iii) the verbatim English prompt templates for both stages. All numbers are produced by `analysis/analyze.py`.

1 Reproducibility and computational details

We simulate the public-goods game with costly peer punishment of Herrmann *et al.* [1] on the FAIRGAME framework [2]. Four agents are matched as partners for ten periods. Each period every agent receives a 20-token endowment and a contribution prompt, and returns reasoning terminated by `CONTRIBUTION = X` ($0 \leq X \leq 20$); in treatment P a second prompt elicits a punishment vector `DEDUCT: A=a B=b C=c` ($0 \leq a, b, c \leq 10$). Co-players are re-labelled every period to block identity tracking, and decisions within a stage are gathered in one lockstep batch.

Each model is served with vLLM at temperature 0.8 using a fixed random seed (`20080307`); all runs were executed offline on a single NVIDIA RTX PRO 6000 (96 GB). The design crosses two treatments (N, P) with a language block (five languages, neutral disposition), a disposition block (four dispositions, English) and a society block (sixteen neutral city personas, English), ten independent groups per cell, sharing the English–neutral–no-society cell; this yields 480 games per model and 1440 in total. Punishment is classified by the contribution deviation between target and punisher: deviation ≥ 0 (the target gave at least as much as the punisher) is antisocial, deviation < 0 is punishment of free-riding. The format non-compliance rate is logged for every cell.

Table S1: Models evaluated. All are open-weight instruction-tuned checkpoints served identically via vLLM at temperature 0.8.

Short name	Hugging Face identifier	Reference
Qwen2.5-7B	`Qwen/Qwen2.5-7B-Instruct`	[3]
Gemma-2-9B	`google/gemma-2-9b-it`	[4]
Llama-3.1-8B	`meta-llama/Llama-3.1-8B-Instruct`	[5]

2 Full statistical tables

Table S2 reports the punishment effect on cooperation (P−N) with Hedges g and rank-biserial r effect sizes, Holm-adjusted Mann–Whitney p-values and 95% confidence intervals: only Gemma differs from zero, and in the negative direction ($g = -2.93$, $r = -0.94$). Table S3 gives the full deviation-binned punishment for all three models, showing that only Gemma reproduces the human monotone free-riding gradient. Table S4 reports the cross-societal correlations with Fisher-z confidence intervals; none approaches the human $\rho \approx -0.90$. Table S5 gives the full per-society contributions for the primary model against the human pools, and Table S6 the dispositional breakdown, in which a *vengeful* prompt raises the antisocial share without restoring the institutional function of punishment.

Table S2: Punishment effect on cooperation with effect sizes: Hedges g and rank-biserial correlation r for the P−N contrast on the ten per-group means, with Holm–Bonferroni adjusted Mann–Whitney p-values and 95% confidence intervals. Human grand-mean gain ≈ +4.3 tokens [1].

Model	Δ (P−N)	Hedges g [95% CI]	r	MWU p (Holm)	Antisocial
Qwen2.5-7B	-0.36	-0.22 [-1.06, 0.62]	-0.11	0.705 (1.000)	59%
Gemma-2-9B	-3.55	-2.93 [-4.20, -1.66]	-0.94	0.000 (0.001)	55%
Llama-3.1-8B	+0.08	0.06 [-0.78, 0.90]	+0.04	0.910 (1.000)	54%

Table S3: Mean punishment expenditure by contribution deviation (target minus punisher) for the English, neutral baseline, treatment P. Deviations < 0 are punishment of free-riding; deviations ≥ 0 are antisocial.

Model	≤ -11	-10 to -1	0	1 to 10	11 to 20
Qwen2.5-7B	2.78	2.42	2.29	2.40	2.15
Gemma-2-9B	3.71	2.45	1.40	2.17	2.06
Llama-3.1-8B	2.84	2.86	2.60	2.78	2.53

3 Prompt templates (English)

The verbatim English contribution-stage and punishment-stage prompts are reproduced below; the four non-English versions are token-for-token translations preserving all placeholders, available in the project repository. Placeholders in braces are substituted at run time; {SOCIETY_BLOCK} and {PERSONALITY_BLOCK} are empty in the neutral, no-society condition.

Contribution stage

```
You are {PLAYER_ID}, one of {N_PLAYERS} members of a group. The same {N_PLAYERS}
members stay together for the whole game.

The game lasts {N_ROUNDS} periods. At the start of EVERY period each member
receives a fresh endowment of {ENDOWMENT} tokens. In each period you privately and
simultaneously decide how many of your {ENDOWMENT} tokens to put into a shared
group project; you keep the rest in your private account.

How tokens become income each period:
- Every token you keep in your private account is worth 1 token of income to you.
- Every token placed in the group project is shared among all {N_PLAYERS} members:
  each member receives {MPCR} tokens of income for EACH token in the project, no
  matter who put it there.

You contribute {ACTION_DESC} to the group project this period.

The other members are shown to you only under temporary labels ({OTHER_LABELS}).
These labels are reassigned every period and do not tell you who is who across
periods. You cannot communicate in any way, and all decisions are anonymous.
{TREATMENT_BLOCK}
After each period you are shown how many tokens each member placed in the project
that period, the total in the group project, and your own income for that period.
{SOCIETY_BLOCK}
{PERSONALITY_BLOCK}
CURRENT SITUATION
This is period {CURRENT_ROUND} of {N_ROUNDS}.

Results of the periods played so far:
{HISTORY_BLOCK}

YOUR DECISION
Decide how many of your {ENDOWMENT} tokens to place in the group project this
period ({ACTION_DESC}). You may reason briefly first. Then finish your reply with
```

Table S4: Cross-societal structure (treatment P, 16 society personas) with Fisher-z 95% confidence intervals. Human reference: Spearman(antisocial, cooperation) ≈ -0.90 [1].

Model	$\rho_{\text{anti,coop}}$ [95% CI]	$\rho_{\text{LLM,HUM}}$ [95% CI]	Wilcoxon p	$\overline{\text{LLM} - \text{HUM}}$
Qwen2.5-7B	-0.09 [-0.56, +0.43]	+0.33 [-0.20, +0.71]	0.669	+0.80
Gemma-2-9B	-0.15 [-0.60, +0.37]	+0.06 [-0.45, +0.54]	0.005	-3.30
Llama-3.1-8B	+0.30 [-0.23, +0.69]	+0.17 [-0.35, +0.62]	0.025	-2.47

Table S5: Per-society contributions for Gemma-2-9B against the human pools of Herrmann *et al.* [1]; antisocial = total antisocial punishment points (treatment P). Societies ordered by human treatment-P contribution.

Society	LLM N	LLM P	Antisocial	Human N	Human P
Boston	13.52	9.70	1487	9.3	18.0
Copenhagen	13.00	9.05	1404	11.5	17.7
St.Gallen	13.97	9.78	1460	10.1	16.7
Zurich	13.33	9.50	1496	9.3	16.2
Nottingham	13.12	9.49	1446	6.9	15.0
Seoul	13.44	9.57	1346	7.9	14.7
Bonn	12.97	9.97	1293	9.2	14.5
Melbourne	13.19	9.73	1393	4.9	14.1
Chengdu	13.41	8.94	1374	8.0	13.9
Minsk	12.34	9.70	1589	10.5	12.9
Samara	12.60	9.81	1507	9.7	11.7
Dnipropetrovsk	13.73	9.87	1470	10.6	10.9
Muscat	13.90	9.42	1472	10.0	9.9
Istanbul	13.64	9.77	1218	5.4	7.1
Riyadh	13.46	9.46	1477	7.6	6.9
Athens	13.90	9.36	1592	6.4	5.7

```
EXACTLY one line in this format, and nothing after it:
CONTRIBUTION = X
(where X is an integer from {CONTRIB_MIN} to {CONTRIB_MAX})
```

Punishment stage (treatment P)

```
You are {PLAYER_ID}. This is the second stage of period {CURRENT_ROUND} of
{N_ROUNDS}.

This period you placed {MY_CONTRIB} tokens in the group project. The total placed
by all members was {GROUP_PROJECT_TOTAL}, and your income from the first stage
this period is {MY_INCOME_THIS_PERIOD} tokens.

Here is how much each of the other members placed in the project this period:
{OTHERS_BLOCK}

You may now assign deduction points to each of these members, simultaneously and
anonymously. For EACH member you choose a whole number of deduction points from 0
to {MAX_PUNISH}.
- Each deduction point you assign costs YOU {PUNISH_COST} token.
- Each deduction point you assign reduces THAT member's income this period by
  {PUNISH_IMPACT} tokens.
The other members decide at the same time and may assign deduction points to
anyone, including you. No one is told who assigned points to whom; you are only
told the total number of points you received.
{RECEIVED_LAST_PERIOD}
{SOCIETY_BLOCK}
{PERSONALITY_BLOCK}
```

Table S6: Dispositional steerability (English, treatment P): mean contribution and antisocial share of punishment by disposition prompt and model.

Model	Disposition	Mean contribution	Antisocial share
Qwen2.5-7B	Neutral	13.90	59%
	Cooperative	16.70	64%
	Selfish	8.43	63%
	Vengeful	12.28	63%
Gemma-2-9B	Neutral	9.86	55%
	Cooperative	14.28	61%
	Selfish	6.41	59%
	Vengeful	8.39	66%
Llama-3.1-8B	Neutral	10.75	54%
	Cooperative	11.95	57%
	Selfish	8.40	56%
	Vengeful	10.70	56%

```
DEDUCTIONS SO FAR
{PUNISH_HISTORY_BLOCK}

YOUR DECISION
Decide how many deduction points (0 to {MAX_PUNISH}) to assign to each member
shown above. You may reason briefly first. Then finish your reply with EXACTLY one
line in this format, and nothing after it:
DEDUCT: A=a B=b C=c
(where a, b, c are integers from 0 to {MAX_PUNISH}, for Member A, Member B,
Member C)
```

References

[1] Benedikt Herrmann, Christian Thöni, and Simon Gächter. "Antisocial punishment across societies". In: *Science* 319.5868 (2008), pp. 1362–1367. DOI: `10.1126/science.1153808`.

[2] Alessio Buscemi et al. "FAIRGAME: a Framework for AI Agents Bias Recognition using Game Theory". In: *ECAI 2025 — 28th European Conference on Artificial Intelligence*. Vol. 413. Frontiers in Artificial Intelligence and Applications. IOS Press, 2025, pp. 4097–4104. DOI: `10.3233/FAIA251300`.

[3] Qwen Team. *Qwen2.5 Technical Report*. arXiv:2412.15115. 2025. arXiv: `2412.15115 [cs.CL]`.

[4] Gemma Team. *Gemma 2: Improving open language models at a practical size*. arXiv:2408.00118. 2024. arXiv: `2408.00118 [cs.CL]`.

[5] Llama Team. *The Llama 3 herd of models*. arXiv:2407.21783. 2024. arXiv: `2407.21783 [cs.AI]`.